



Sustainable Hospital Flagship (SHF): Transition from waste to value



General information

1.a Applicants

Table 1 : Applicants

Name	Institution	Department	Email	Tenured (i.e. fixed) position/ Tenure-track/ Other (please specify)	For Erasmus MC applicants: clinician/ researcher/ mix	Role in Flagship [Main Lead/Lead/ Applicant]
Dr. Nicole Hunfeld	Erasmus MC	ICU/PHARM	n.hunfeld@erasmusmc.nl	Tenured	mix	Main lead & WP lead
Prof. Dr. Diederik Gommers	Erasmus MC	ICU	d.gommers@erasmusmc.nl	Tenured	mix	Applicant
Prof. Dr. Dick Tibboel	Erasmus MC	ICU	d.tibboel@erasmusmc.nl	Tenured	researcher	Applicant
Maarten Timmermann	Erasmus MC	Buyer group	m.timmermann@erasmusmc.nl	NA	NA	Applicant
Dr. Bas van Rijn	Erasmus MC	OBS	b.vanrijn@erasmusmc.nl	Tenured	mix	Applicant
Dr. Roelof van Leeuwen	Erasmus MC	ONC/PHARM	r.w.f.vanleeuwen@erasmusmc.nl	Tenured	mix	Applicant
Dr. ir. JC Diehl	TU Delft	IDE	j.c.diehl@tudelft.nl	Tenured		Lead & WP Lead
Dr. Saba Hinrichs-Krapels	TU Delft	TPM	s.hinrichs@tudelft.nl	Tenure-track		Applicant
Prof. Dr. ir. Conny Bakker	TU Delft	IDE	c.a.bakker@tudelft.nl	Tenured		Applicant
Prof. Dr. ir. Ruud Balkenende	TU Delft	IDE	a.r.balkenende@tudelft.nl	Tenured		Applicant
Dr. ir. Jotte de Koning	TU Delft	IDE	jotte.dekoning@tudelft.nl	Tenure-track		Applicant
Prof. Dr. ir. Erik van Raaij	Erasmus University Rotterdam	RSM-TOM ESHPM-HSMO	eraaij@rsm.nl	Tenured		Lead & WP Lead
Prof. Dr. Pilar Garcia Gomez	Erasmus University Rotterdam	ESE-Applied Economics	garciagomez@ese.eur.nl	Tenured		Applicant
Dr. Sophie van der Zee	Erasmus University Rotterdam	ESE -Applied Economics	vanderzee@ese.eur.nl	Tenured		Applicant & WP Lead
Prof. Dr. Albert Wagelmans	Erasmus University Rotterdam	ESE - Econometrics	wagelmans@ese.eur.nl	Tenured		Applicant & WP Lead
Dr. Wilco van den Heuvel	Erasmus University Rotterdam	ESE - Econometrics	wvandenheuvel@ese.eur.nl	Tenured		Applicant

Corresponding Lead: Dr. Nicole Hunfeld

Planned start date of seed funding: 01/09/2022

Primary Convergence theme: *Technology Supported Transitions in Health(care)*

Keywords: *Environmentally Sustainable Hospital; Circular Healthcare; Sustainable Healthcare; Circular Transition, Technology Transition*

Partners: Metabolic, Philips, Fresenius Medical Care, PreZero, MINT Zorgadvies, MVO Nederland, NEVI, Gemeente Rotterdam

1.b Summary (max 300 words)

The healthcare sector is one of the most carbon-intensive sectors, contributing to 7% of emissions in the Netherlands. In response, the Dutch government has called for more environmentally sustainable healthcare and a 49% emission reduction by 2030. The Sustainable Hospital Flagship (SHF) addresses this issue with its **mission** to shape the future of healthcare in a transformative way by introducing environmentally sustainable healthcare practices to reduce the sector's ecological footprint. Our **aim** is to build a strong research community to create an evidence-based transition model for environmentally sustainable hospitals using the expertise of three Dutch research institutions. This interdisciplinary team will improve or maintain current healthcare practices while minimizing negative environmental impacts. The project connects researchers with expertise in engineering, design and policy (TU Delft); medical sciences (Erasmus MC); operations & technology management, and social sciences (Erasmus University Rotterdam).

In five years SHF aims to:

- Publish an evidence-based transition model for environmentally sustainable hospitals;
- Research and publish at least nine evidence-based cases within three clinical departments: Intensive Care Units, Oncology and Obstetrics;
- Publish scientific articles in the fields of design, economics, medical & environmental sciences and contribute to professional journals for clinicians, nurses, purchasers, supply chain managers & other healthcare practitioners;
- Develop a strong interdisciplinary Convergence community;
- Create a physical space, the Sustainable Hospital Field Lab at Erasmus MC, and an online platform, (<https://circle.so>), to develop, co-create and learn from each other;
- Educate at least 5 PhDs, 3 postdocs, 20-30 master students from at least four educational programs within three academic institutions;
- Develop circular innovations with the potential for spin-off companies or adoption by industry partners;
- Disseminate project results through our website (www.sustainablehospital.org) and podcast *The Sustainable Hospital*, social media, television/radio appearances, an annual conference and newspaper publications.

1.c Public Summary (max 300 words)

Although the healthcare sector aims to restore, protect, and improve the health of society, it is also responsible for creating large amounts of waste, pollution, and carbon emissions. These factors all contribute to climate change, which in turn can contribute to health problems. In the Netherlands, health care is responsible for 7% of the CO₂ footprint. Our research **mission** is to solve this urgent societal issue by enabling hospitals to reduce their ecological footprint. We will do this by bringing together researchers with a wide range of knowledge who will work together to develop new and improved technologies and methods that will reduce the environmental impact of healthcare. Experts from three Dutch institutions will be involved in this project: TU Delft, Erasmus University Rotterdam, and Erasmus MC. Our **aim** is to build a strong research community that will create positive impacts on science and society. We will create impact by publishing our findings, educating professionals and introducing new innovations into healthcare practice. We will involve essential stakeholders to co-develop new designs, form coalitions and embed the solutions into the healthcare system. We also want to investigate the impact of our interventions on the environment, costs, the supply chain, behavior and policy. We see this project as a steppingstone to reach our final goal: a 100% carbon-neutral Erasmus MC in 2030. We will advocate for an environmentally sustainable future using the project outcomes and the societal traction they create. We will enable other hospitals at the regional and international level to adopt our solutions so that the sustainable healthcare transition can spread like wildfire. We will reach out to the public in various ways by telling our story on our podcast, *The Sustainable Hospital*, by media appearances and through our website www.sustainablehospital.org.

2.Relevance to Convergence Health & Technology mission (50%) — max. 2500 words for 2.a to 2.e

2.a Contribution to H&T mission and fit with H&T theme(s)

There is global recognition that the healthcare sector is one of the most carbon-intensive sectors, accounting for 4.4% of global net greenhouse gas emissions and toxic air pollutants (1). In the Netherlands alone, 7% of the national carbon footprint is associated with the healthcare sector (2). The daily material consumption of hospitals is enormous through the use of disposable products such as syringes, medication, dressings, catheters, and personal protective equipment. This negatively affects climate change, depletes limited natural resources, and adversely impacts global population health. Many organizations and researchers have made a call to action to improve the ecological impacts of healthcare, including the recent ICCP report (3), the World Health Organization (WHO)(4, 5), and medical journals (6). The UN Sustainable Development Goals (SDGs) also identify a healthy society and planet as priorities (SDG3, SDG 12 and SDG 13).

The goal of the Sustainable Hospital Flagship (SHF) is to reduce healthcare's ecological footprint and also contributes to the H&T missions of *forming novel frameworks that foster scientific discovery and technological innovation*. The **mission** of SHF is to shape the future of healthcare in a transformative way by introducing environmentally sustainable healthcare practices to reduce healthcare's ecological footprint. Our **aim** is to build a strong research community to create an evidence-based transition model for environmentally sustainable hospitals using the expertise of three Dutch research institutions and our partners. We will improve or maintain current healthcare practices while minimizing negative environmental impacts.

The SHF also fits within the “technology-supported transition in healthcare theme” and its mission to *create a resilient, value-based, and affordable healthcare system*. The Green Deal for Sustainable Healthcare, established by the Dutch government, targets a 49% reduction in healthcare-related carbon emissions by 2030 from 1990 levels (7, 8). By 2050, all healthcare organizations must be carbon neutral. Our ambition is to accelerate this transition by conducting interdisciplinary research to develop new and improved technologies and methodologies such as tools, models, devices, and assessment methods that enable healthcare practices that reduce adverse ecological impacts. Of key importance is that these are developed in close collaboration with stakeholders within and outside this consortium. As described by the WHO (5), healthcare transitions will only be successful through creating awareness among healthcare professionals, involvement of key societal stakeholders in and outside the healthcare sector, community engagement, and support of the local and national governments.

An interdisciplinary way of working is also at the core of this entire project. This approach ensures that the developed technologies and methodologies meet the end-users' needs, the time-to-market is reduced, early and effective adoption is realized, all of this ultimately stimulating wide implementation of project outcomes in the healthcare system. This project will shape what the healthcare system of the future will look like with a focus on creating an understanding of the role new technologies play in making the healthcare sector more environmentally sustainable, and developing new methods to evaluate the environmental and societal impact of these technologies.

We aim to strengthen the impact of Convergence Health & Technology in the region of South Holland, the Netherlands and internationally by wide dissemination of our results and learnings to both an expert and layman audience. We will concurrently educate the next generation of professionals by equipping them with transdisciplinary skills and competences that are required to implement

breakthrough innovations supporting environmentally sustainable healthcare practices now and in the future. This consortium aims to become the global leader in this field by creating a vibrant ecosystem of researchers, students, clinicians, and entrepreneurs that interact, inspire, co-create, and support each other.

2.b Synergy between disciplines

The transition to environmentally sustainable healthcare practices is complex and requires input from multiple perspectives to be implemented successfully. Therefore, we will use an interdisciplinary approach that allows for blending and integration of multiple knowledge pools. This project will combine the expertise from engineering, design, and policy (TU Delft), medical sciences (Erasmus MC), operations & technology management, and social sciences (Erasmus University Rotterdam, EUR). This wide range of knowledge allows us to collaboratively design circular interventions by including insights from clinical practice, industrial design, behavioral economics, supply chain management, and operations research.

This project consists of three phases: 1) Defining the circular cases and assessment model, 2) Design & optimization of technologies and methodologies and 3) Integration, dissemination, and implementation. Each phase requires the combined expertise from the three participating institutions, consortium scientists and societal partners. For example, the circular cases (Phase 1) will be defined based on input from clinicians (Erasmus MC) that are familiar with the healthcare context together with scientific evidence and expertise of societal partners in choosing promising avenues for product and technology development (TU Delft) and processes and control of behavior (EUR and TU Delft). During Phase 2 there will be a strong interaction and integration of design engineering, operations research, supply (chain) management, and behavioral science to optimize the designs of technology and methodologies that are developed for the transition towards an environmentally sustainable healthcare system. These joint efforts will allow distinct disciplines to merge their approaches as they will need to rely on each other during this project. The societal partners will function as an advisory group during Phase 2 and will be strongly encouraged to cooperate and co-create during Phase 3.

As a result of the energy coming out of the pre-flagship meetings held by this consortium in the past year, synergies and collaborations have already emerged. Six master students from TU Delft and EUR are already working on pilot projects at the Green ICU Room (9). We plan to extend the current Green ICU room into a Sustainable Hospital Field Lab, where we offer our joint expertise to advise other parties on matters related to the program. Green teams from Erasmus MC and from regional hospitals (through Klimaattafel Rotterdam), societal and industrial partners will be involved in this field lab to share knowledge easily. Additionally, we will build an online community platform (circle.so) that will be partly open to the public, and will house closed communication platforms for students, staff, and consortium members to share insights and foster additional collaboration among all disciplines involved in this project. This community will be unique in its interdisciplinary character and connecting students, early-career researchers with junior and senior healthcare professionals, policymakers, politicians, and board members that we foresee to be very valuable far beyond this project.

2.c Long term sustainability plan

As stated in the introduction, there is widespread recognition regarding the urgency in reducing the environmental impacts of the healthcare system. Given Dutch and international climate targets by 2030 and various initiatives related to sustainable healthcare, we expect a lot of traction and support in this field. Interest in sustainable healthcare transitions is rapidly increasing (3-6), opening new funding opportunities on national, European, and industrial levels and we expect more to come. Due to its complexity, the transition to environmentally sustainable healthcare practices will take time. We

see our project as a seed that we plant that will require years to fully mature but which will flourish once it matures to its full potential. This is like what we observed in other impactful transitions within healthcare, such as the implementation of the patient safety programs that took 10 years. With the support of this Flagship Program, the SHF consortium will be able to develop as a leading international research hub on this theme to create a solid basis to attract larger funding in the coming years.

One potential project risk is another epidemic or political tension that will inhibit easy access to clinical practice within the Erasmus MC, which may impact the Sustainable Hospital Field Lab, end-user interaction and PhD research and master projects. Therefore, we will facilitate our community building both offline and online through the circle.so platform. This will ensure all participants in this project can collaborate and share knowledge in any scenario. We will use the experiences gained by all consortium members in educating and supervising students during the lockdowns in the past 2 years.

Another potential risk is a conflict between one or two consortium partners, as this will significantly impact the project outcomes. To build trust and facilitate communication, the whole consortium will meet online at least bi-weekly. Every quarter, there will be a physical meeting within the Sustainable Hospital Field Lab to provide the opportunity to discuss the progress of the program. In case of difficulties, the main Lead (Nicole Hunfeld) and the program secretary will be the contact points for reporting. They are aware of internal experts within the three institutions to consult for legal assistance and mediation.

2.d Visibility, attracting talents and new partners

Since the start of our work within the Green ICU Room, we have noticed that this topic receives a lot of attention from the media (10-16), politicians, and practitioners. For example, Diederik Gommers discussed sustainable healthcare with *Humberto Tan* (June 6th 2021) and at *Een Vandaag* (October 4th 2021), Nicole Hunfeld joined *Hart van Rotterdam* (October 26th 2021) and *Atlas* (October 6th 2021) and in the Dutch newspaper *the Volkskrant* (13) on September 15th 2021, and JC Diehl and Nicole Hunfeld presented at the opening of the Dutch Intensivist Days 2022. Additionally, two Dutch Ministers (Van Engelshoven in 2021 and Dijkgraaf in 2022) have visited the Green ICU at Erasmus MC. We will use this existing traction for this project; we envision the cases that we will work on to be very attractive for the media to report about. This will be supported by the dissemination of project results through our website (www.sustainablehospital.org), social media, television/radio appearances, newspapers, and scientific publications. Additionally, we aim to join podcasts such as Medisch Contact, Mednet, and BNR to discuss the topic of sustainability within healthcare. The topic of sustainability in healthcare has also received increasing attention from students and researchers. We already started with the first pilot students from TU Delft and EUR (17) and they were asked to show their results to the Ministry of Education twice. Also, medical device and pharmaceutical companies (i.e., Johnson and Johnson, Fresenius Medical Care, Philips, and Merck) are looking for sustainable solutions and are approaching the SHF already for pilot projects.

We have noticed that the strong interdisciplinary convergence community that SHF is building is attractive for students and young researchers to join because of the urgency young people feel to work on a transition like this to save our planet for our and future generations. The wide range of research domains, experts, and the physical space that we are creating within the Erasmus MC has already shown to be instrumental in providing an inspiring and attractive workplace during its pilot phase. This Sustainable Hospital Field Lab will be of huge importance to provide a physical space where people can meet and conduct research. Not all actors involved in this project will be in the position to travel to the Erasmus MC on a regular basis, to facilitate the online community building, we envision we will set up an innovative online community platform (circle.so) to interact, inspire and connect

with peers of all three institutions. This will be a unique way to network with academic and societal partners and to gain fast input and feedback on research output.

So far, this project is supported by the granted project Digital Health in Circular Economy (DICE) funded by Horizon Europe and Circular Plastic Medical. Several other grant applications are submitted or will be submitted soon: ZonMW open competition, NWA-ORC route on Circular Economy, CO FUND (Horizon Europe) on inclusive sustainability, NWO Raak, Alliance Zero, Groeifonds and a European sustainability call and ERASMUS+ in 2022.

2.e Long term societal impact

Table 2 presents the impact pathways this project will entail on science and education, healthcare, and society.

Table 2 : Pathways to impact

	Impact on Science and Education	Impact on Healthcare	Impact on Society
Research community & activities & Sustainable Hospital Field Lab	- Increased knowledge by scientific publications in peer-reviewed journals - Increased advocacy & engagement by targeted engagement of stakeholders by a monthly webinar - Knowledge sharing and networking by the annual conference organized by the consortium	- Assisting healthcare professionals at Erasmus MC to have a positive impact - Assist implementation of evidence-based practices - Impact on healthcare policymakers and healthcare payers by sharing knowledge - Adaptation by Klimaattafel Rotterdam	Impact on patients and citizens to be treated and live in a healthier environment
Evidence-based transition model & 9 cases and best-practices	- Impact on the purchasing profession, especially purchasing in healthcare - Impact on the design of technologies supporting environmentally sustainable healthcare - Increasing knowledge on behavioral change to support transitions in healthcare	- Improving the purchasing function within hospitals by providing them with evidence-based strategies - Providing information on new technologies and other innovations to suppliers, not only first-tier suppliers but also further upstream - Valorization of innovations resulting in new companies	Impact on the city of Rotterdam and supporting them in the ambition to develop a more sustainable city, and impact on the region (province of South Holland) and the Netherlands
Educational practices	Impact on students in multiple programs at Erasmus MC, TU Delft and EUR		
Dissemination	Use of our cases and experiences in education	Reaching a large group of healthcare professionals:	Reaching a large audience to advocate

	and through publication of data	sharing knowledge through Klimaattafel Rotterdam and on a (inter)national level	for this topic
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3. Flagship proposal (25%) max 2000 words for 3.a to 3.d

3.a Ambitions

SHF and its partners have the **mission** to shape the future of healthcare in a transformative way by introducing environmentally sustainable healthcare practices supported by existing and emerging technology, to reduce healthcare's ecological footprint. Our **aim** is to build a strong research community to create an evidence-based transition model for environmentally sustainable hospitals using the expertise of three Dutch research institutions, whereby we improve or maintain current practices while minimizing negative impacts on the environment.

3.b Aims and feasibility

In five years SHF aims to:

- Publish an evidence-based transition model for environmentally sustainable hospitals;
- Research and publish at least nine evidence-based cases within three clinical departments: Intensive Care Units, Oncology and Obstetrics;
- Publish scientific articles in the fields of design, economics, medical & environmental sciences and contribute to professional journals for clinicians, nurses, purchasers, supply chain managers & other healthcare practitioners;
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3.c Work plan

Table 3: Overview of the workplan

Work Packages	WP Leaders & Institution (own investment)	Funded by Convergence	Duration/start of the work
WP 1 Hospital Workflow and Impact Analysis	Hunfeld (EMC) Diehl (TUD) Van Rijn (EMC) Garcia Gomez (EUR)	-Postdoc (2.5 year) -PhD	-starts in year 1 -starts in year 1
WP 2 Sustainable Hospital Field Lab	Gommers (EMC) Diehl (TUD) Bakker (TUD) Hunfeld (EMC)	-PhD -PhD	-starts in year 1 -starts in year 2
WP 3 Behavioral Analysis for Sustainable Interventions	Van der Zee (EUR) Garcia Gomez (EUR) Diehl (TUD)	-PhD -Research Asst (1 year)	-starts in year 1 -in first year of project
WP 4 Purchasing and Supply Management for the Sustainable Hospital	Van Raaij (EUR) Hinrichs-Krapels (TUD) Timmermann (EMC)	-PhD -Research Asst (1 year)	-starts in year 2 -in first year of project
WP 5 Operations Research for Optimizing Material Flows and Systems	Wagelmans (EUR) Van den Heuvel (EUR) Van Leeuwen (EMC) Balkenende (TUD)	-Postdoc (5 year) -Research Asst (1 year)	-starts in year 1 -in first year of project
WP 6 Environmentally Sustainable Healthcare Transition Framework	Diehl (TUD) Van Raaij (EUR) Hunfeld (EMC) de Koning (TUD)	-Postdoc (2.5 year)	-starts in year 3
WP 7 Project Management, Education and Dissemination	Hunfeld (EMC) Van Rijn (EMC) Van Raaij (EUR) Diehl (TUD)	-Community manager -Program secretary	-all five years -all five years

This unique project (Table 3) assesses the environmental impacts of three healthcare departments: the ICU (Adult, Pediatric and Neonatal), Oncology and Obstetric Care and will propose technology-supported solutions for the transition towards environmentally sustainable healthcare practices by taking a holistic approach. This project consists of seven work packages (WPs) (Figure 1), led by senior scientific staff from the three academic institutions that closely interact with and rely on each other's input and outcomes. Overall we will use circular economy principles to develop environmental sustainable (technology) transitions.

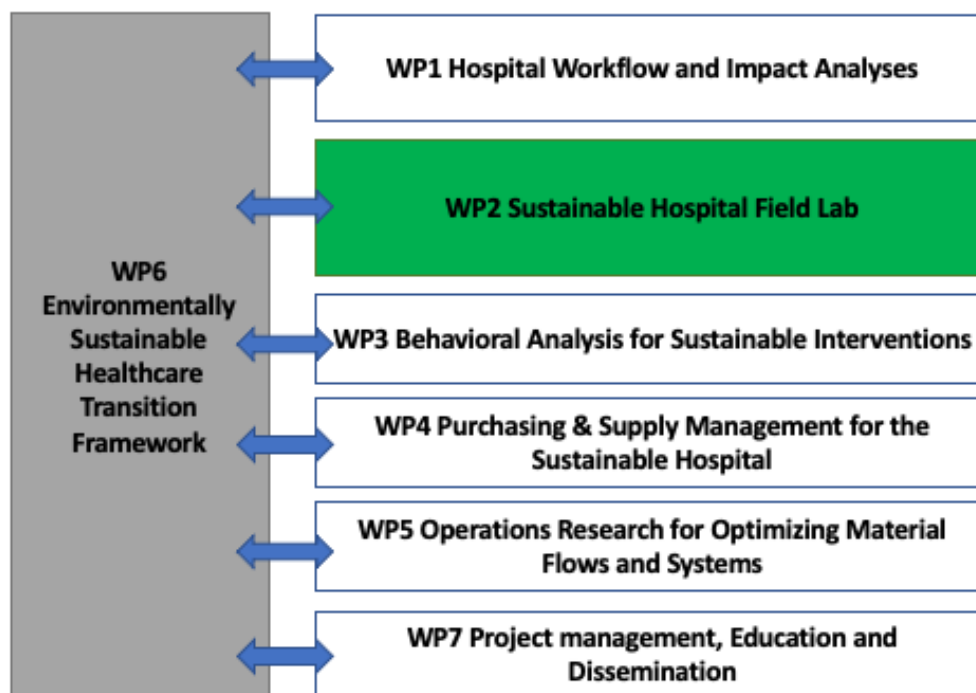


Figure 1: Overview of how the work packages are connected

WP 1 Hospital Workflow and Impact Analysis

WP lead: Nicole Hunfeld (Erasmus MC)

The main aim of WP 1 is to define a Sustainable Hospital (SH) assessment method and to select nine circular intervention cases that will be the input for WP 2, 3, 4, 5 and 6.

Part 1. Definition of Sustainable Hospital (SH) Assessment Method

T1.1 Development of an SH assessment method. The SH assessment method will be used to quantify the environmental and other relevant impacts of the 9 cases before and after intervention.

Part 2. Assessment of three hospital workflows and best practices

T1.2 The three hospital workflows (ICU, Oncology, and Obstetric Care) will be first assessed by a Material Flow Analysis (MFA) to get an understanding of the main material flow inputs and outputs. These MFA's are already under development by an external agency (Metabolic).

T1.3 A scientific literature review (evidence-based) will be carried out to map what has already been done (best practices).

T1.4 Interviews with staff and suppliers (practice) to identify sustainability challenges in their daily workflow and supply chain.

T1.5 Based on tasks 1.2-1.4 a selection of 10 hotspots for each of the three hospital workflows will be shortlisted.

Part 3. Selection and definition of nine circular intervention cases

T1.6 For each hospital workflow, at least one circular intervention case per product category (disposables, devices & durable equipment) will be selected (minimum of 9 cases) (See Figure 2).

Flagship Contracted staff: 1 Postdoc and 1 PhD

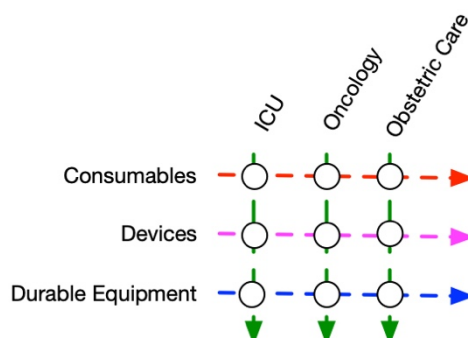


Figure 2: Selection of nine circular intervention cases.

WP 2 Sustainable Hospital Field Lab

WP lead: JC Diehl (TUD)

The Sustainable Hospital Field Lab will be a real-life testing site where various parties collaborate to develop, test, learn to implement and scale-up new technologies for commercial applications. The main purpose of WP 2 is to develop nine environmental sustainability cases.

Part 1. Develop circular product and technology interventions

T2.1 To develop, test, and iterate design solutions for circular interventions based upon the 9R Circular principles (18).

T2.2 Identification, further development, and active integration of emerging technologies (technology roadmap) to foster the environmentally sustainable transition.

Part 2. Stakeholder engagement and co-creation

T2.4 Co-creation and co-validation of the nine cases within the Sustainable Hospital Field Lab with the stakeholders and monthly engagement meetings (online).

Part 3. Development of Circular Design approach

T2.3 The development and validation of a Circular Design approach specifically for medical consumables, devices, and equipment based on experiences in the Sustainable Hospital Field Lab and previous by TU Delft developed more generic Circular Design approaches (19-23).

The nine case studies provide input for WP 3, 4, 5 and 6. In return, these work packages will have input on the development of these cases.

Flagship contracted staff: 2 PhDs

WP 3 Behavioral Analysis for Sustainable Interventions

WP lead: Sophie van der Zee (EUR)

The main aim of WP3 is to determine which factors motivate and hinder sustainable behavior amongst hospital personnel and to utilize these insights to elicit behavioral change.

Part 1. Behavioral analysis

T3.1 Individual interviews with nurses, doctors, management, cleaning staff, and patients in ICUs, obstetrics, and oncology to identify potential drivers and barriers of environmentally sustainable healthcare practices.

T3.2 Focus groups with employees from different professions to exchange and test insights gathered from the interviews.

T3.3 Testing the drivers and barriers identified during the qualitative research in a systematic manner using vignettes (24).

Part 2. Behavioral change

T3.4 Insights from Part 1, in combination with insights from WP1 and 2, will be utilized in four field experiments to test the most promising interventions. To determine the external validity of our results, one intervention will be tested across two departments.

Flagship contracted staff: 1 PhD and 1 Research Assistant

WP 4 Purchasing and Supply Management for the Sustainable Hospital
WP lead: Erik van Raaij (EUR)

Main aim of the WP4 is to design Purchasing and Supply Management (PSM) strategies for enabling circularity.

Part 1. PSM strategies for circularity

T4.1 Map current PSM processes for consumables, devices, and durable equipment, for ICU, Oncology and Obstetric care (25).

T4.2 Identify key purchase categories with the highest impact to become more circular (based on 9Rs). Take inputs from WP 1 and 2; hotspot analysis, outcomes from the field lab.

T4.3 Develop internal demand management strategies per category that create “optimal” incentives for internal users to become more circular. Take inputs from WP 3 on behavioral analysis. Collaborate with WP 5 to model trade-offs between cost, health outcomes, and sustainability.

T4.4 Develop PSM strategies per selected category that create “optimal” incentives for suppliers to contribute to circularity (26).

Part 2. Innovation strategies with suppliers for circularity

T4.5 Systematic review of business models / buyer-supplier relationship models that create incentives for circularity.

T4.6 Co-design PCP/PPI strategies for a limited number of high-impact product categories, ideally spread across the three departments/product types.

Flagship contracted staff: 1 PhD + 1 Research Assistant

WP 5 Operations Research for Optimizing Material Flows and Systems
WP lead: Albert Wagelmans (EUR)

The main aim of WP5 is to use techniques from Operations Research to optimize interventions designed in WP2 and to support WP4 and WP1.

Part 1. Optimization of logistical processes

T5.1 Analysis of current logistical processes associated with the hotspots identified in WP1 to explore opportunities for waste reduction by optimizing their efficiency.

T5.2 Analysis of the logistical processes induced by the interventions proposed in WP2 to identify opportunities for optimization of their effectiveness and/or efficiency.

T5.3 Mathematical and statistical modeling of those processes that lend themselves to optimization.

T5.4 Translation of findings into practical recommendations to stakeholders.

Part2 Decision support

T5.5 Modeling of trade-offs between cost, health outcomes, and sustainability for task T4.3 of WP4.

T5.6 Development of a model/tool to support decisions about the best mix of interventions that can be undertaken when the budget (or other resources) is limited and multiple objectives must be considered.

Flagship contracted staff: 1 Postdoc and Research Assistant (1 year)

WP 6 Environmentally Sustainable Healthcare Transition Framework

WP lead: All three leads (Nicole Hunfeld, JC Diehl, Erik van Raaij)

T6.1 Development of evidence-based transition model for environmentally sustainable hospitals. The model will be constructed by literature review, interviews with transition experts, reflective practice (27-31) on the nine cases of the Sustainable Hospital Field Lab (WP 2) and knowledge gained in WP 4, 5 and 6.

Flagship contracted staff: 1 Postdoc (2.5 years)

WP 7 Project Management, Education and Dissemination

WP lead: Nicole Hunfeld (EMC)

T7.1 Project management and community building by organizing activities for the consortium, partners and broader (scientific) audience. By moderating the circle.so platform, organizing the weekly consortium meetings, the monthly webinars, and the annual conference.

T7.2 Dissemination of results and best practices with a larger audience, for example through our website (www.sustainablehospital.org), social media, our podcast the sustainable hospital, other podcasts, and media appearances.

T7.3 Development of course materials for the online course

Flagship contracted staff: Community manager (0.5 fte) and program secretary (0.5 fte)

At the start of the project, a data repository of one of the three institutions will be chosen to share data efficiently and safe within the entire consortium. All consortium members will ensure that the data they collect is additionally stored in their host institution's data repository.

3.d Alignment of the program with relevant national and international research agendas

This sense of urgency of this consortium to bring this project forward and aligned greatly with national and international research agendas and policies that ask for action, for example:

- The publication of the recent ICCP report (3).

- The WHO pushes its member states by providing various guidelines on the implementation of environmentally sustainable healthcare practices (4, 5).
- It supports the Sustainable Development Goals, in particular: SDG 3, SDG 12 & SDG 13 (32).
- The Dutch Healthcare sector signed the Green Deal 2.0 to reduce the ecological footprint of healthcare drastically by 2030 (8).
- The Dutch Research Agenda (NWA) Route on Circular Economy and Resource Efficiency: sustainable circular impact (33).
- The Horizon Europe Framework Programme (Cluster 1 and Cluster 6) (34).
- Sustainability Agenda Municipality Rotterdam (35).

4. Consortium (25%) max 2000 words for 4.a to 4.d

4.a Track record of the consortium members

The consortium members represent all three institutes, with one lead from each institute. The members represent eight departments/faculties in total (EMC-ICU, EMC-Onco, EMC-Obst, TUD-IDE, TUD-TPM, EUR-ESE, EUR-RSM, EUR-ESHPM).

Leads of the project

Nicole Hunfeld, main lead of this project, is pharmacist at Erasmus MC and the vice-president of KNMP and is a well-known guest at (inter)national conferences about pharmacology and has a broad medical and pharmaceutical network. She chairs the national Green ICU committee.

Erik van Raaij is professor of Purchasing and Supply Management in Healthcare at Erasmus University Rotterdam. His expertise covers all aspects of purchasing and supply management, including issues of supply chain management, in the healthcare sector.

Jan Carel Diehl is associate professor at the section Design for Sustainability at the Faculty of Industrial Design Engineering at TU Delft. He leads the Healthcare for All and Inclusive Circular Development research programs of the Delft Global Initiative (DGI) and is director of the Delft Design Lab (DDL) Inclusive Global Health.

The three leads are supported by excellent consortium members

Erasmus MC

Diederik Gommers is head of the department and professor of Intensive Care. He is the chairman of the Dutch Union for Intensive Care. During the COVID-19 pandemic in the Netherlands, Gommers has been part of the Outbreak Management Team that advises the government of Mark Rutte on measures required to reduce the spread of the disease.

Dick Tibboel is professor and former head of the Pediatric Intensive Care of Sophia Children's Hospital. Between 2012 and 2018 he was director of research within the same department.

Maarten Timmermann is Lead Impact circular and sustainable procurement in the procurement department.

Bas van Rijn is associate professor and obstetrics consultant where he works on implementing the green delivery. He is also a member of the Gyna Goes Green, the national sustainability platform of gynecologists.

Roelof van Leeuwen is assistant professor within the department of clinical pharmacy and oncology. He has an excellent track in effective care in oncology and in optimization of clinical processes.

TU Delft

Conny Bakker is professor in Design for Sustainability and Circular Economy at the Faculty of Industrial Design Engineering. She leads the research group Design for Sustainability, currently 30 persons, and oversees a Circular Design portfolio of projects worth several million euros, with a wide range of (inter)national partners.

Ruud Balkenende is professor of Circular Product Design and director of education at the Faculty of Industrial Design Engineering. His research focuses on design for a circular economy, specifically design approaches for reuse, repair, remanufacturing and recycling of products and materials.

Saba Hinrichs-Krapels is associate professor in health systems and policy and holds the prestigious Delft Technology Fellowship award. Using approaches from implementation science, engineering design and complexity modeling, she explores how theoretical and empirically derived models for improving health systems can be shaped into real interventions that improve health systems. She has experience in applying these approaches in the design of purchasing processes for hospitals.

Jotte de Koning is assistant professor of Design for Sustainability Transitions at the Faculty of Industrial Design Engineering. Jotte conducts research on the intersection of the fields of design for sustainability and systems thinking.

Erasmus University

Sophie van der Zee is assistant professor in Behavioral Economics. She has a background in psychology and computer science. She studies human behavior and decision-making in several contexts, including (in)secure and (dis)honest behavior.

Albert Wagelmans is professor of Management Science. His expertise is in the optimization of transportation and logistical processes through statistical and mathematical modeling. His work includes projects in healthcare and humanitarian logistics, for which he collaborated with various departments of Erasmus MC and with organizations such as the Dutch Transplant Foundation (NTS) and North Star Alliance.

Pilar Garcia-Gomez is professor of Applied Economics, with a focus on health and labor economics. She is an expert in the use of applied microeconometrics and large administrative datasets of individual information to estimate the causal effect of interventions.

Wilco van den Heuvel is associate professor of Operations Research. His expertise is in optimization using techniques such as mathematical modeling and algorithm development with applications in sustainable logistics such as collaborative transportation and production planning with remanufacturing.

Members of the consortium have a significant track record in research excellence and societal impact.

Albert Wagelmans has a track record in translating scientific results into clinical applications: his algorithms have been adopted by the Dutch Transplant Foundation and the Erasmus MC; for this work he received the ERIM Award for Impact on Management Practices in 2016. Dick Tibboel published 1070 peer-reviewed publications in the field of pediatrics, and his work has led to the publication of national guidelines for pain treatment of children in general and specifically postoperative pain and standardization at an EU level of treatment of neonates with Congenital Diaphragmatic Hernia. Sophie van der Zee was the Erasmus representative in the New Scientist competition for research talent of the year in 2018; in 2020, she was amongst the final three candidates for the Klokhuis Wetenschapsprijs, and she attracted considerable media attention with her work on (in)secure and (dis)honest behavior, including two international documentaries and dozens of TV programs and radio interviews. Saba Hinrichs-Krapels has led the design and delivery of two 'policy labs' in the UK that brought together policymakers, practitioners, and researchers to engage on how to bring research closer to scientific impact, in health policy domains. Diederik Gommers, in his capacity of chairman of the Dutch Union for Intensive Care and member of the

Outbreak Management Team had direct impact on national policy; he joined various Dutch talk shows such as Humberto Tan and Jinek.

Members of the consortium have a significant track record in excellent education, including programme (re)design and engagement of pre-experience students in research. Erik van Raaij is the former academic director of the RSM Bedrijfskunde program and teacher of various research methods courses as well as a MOOC on sustainability (36). Diederik Gommers supervised numerous PhD students. Albert Wagelmans is the former director of the Bachelor-Master Program in Econometrics & Management Science and the Bachelor-Master Program in Informatics & Economics, and he is currently involved in the supervision of the “Impact at the Core” education project at EUR; he supervised over 200 students during their master thesis or internship. Wilco van den Heuvel is the Academic Director of the master specialization Analytics and Operations Research in Logistics. Sophie van der Zee is the incoming academic director and the MSc Thesis coordinator of the MSc Behavioral Economics. Pilar Garcia Gomez is the academic director of the master specialization Health Economics at the Erasmus School of Economics. JC Diehl is author of the United Nations Environment (UNEP) Design for Sustainability manual which is used worldwide for teaching.

Moreover, members of the consortium have demonstrated to be able to attract external funding and successfully participate in large externally funded projects. Diederik Gommers attracted over 2 million euros of external grant funding in the last 5 years. Also Dick Tibboel raised over 1.5 million euros of external grant funding in the last five years. Connie Bakker currently co-leads the European H2020 project ReCiPSS, which helps designers and manufacturers understand how collection, remanufacturing and reuse of products can lead to more profitable, resource-efficient, and resilient business practices. Nicole Hunfeld is involved in numerous ZonMW grants and participated in several research study teams in both the pharmacy and the intensive care. JC Diehl is currently co-principal investigator of the NWO project INSPIRED as well as participating in two other NWO projects. Sophie van der Zee led a Police and Science grant on identifying factors driving people’s willingness to report cybercrime victimization. Albert Wagelmans has received a sustainable Logistics Grant from NWO and co-funding from the NS Dutch Railways. Saba Hinrichs-Krapels was the principal investigator of two large UK grants granted by the Health Foundation and the National Institute for Health Research and co-investigator on 5 others.

4.b Expertise of the consortium members in relation to the aims and objectives of the program

The consortium brings all necessary areas of expertise to the table:

- Expertise in clinical practice from Erasmus MC in the ICU, Oncology and Obstetrics. This expertise safeguards a focus on positive outcomes for patients and healthcare professionals. These consortium members also safeguard access to clinical departments which enables the effective functioning of the field lab at the core of this flagship project.
- Expertise in technology and design from TU Delft. Interventions for environmentally sustainable healthcare practices require analysis and redesign of products as well as work processes.
- Expertise in behavioral economics from EUR-ESE. Whether interventions translate into impact often depends on changes in behaviors. This expertise ensures the focus on people as key levers for the transition to circularity.
- Expertise in purchasing & supply (chain) management from EUR-RSM / EUR-ESHM. Suppliers are a source of innovative ideas, and they need to make the transition together with the hospital. This expertise ensures that the supply chain remains aligned on the trajectory to circularity. This is complemented by the expertise in technology adoption in healthcare from

TUD/TPM, which studies the role and introduction of technologies in a hospital setting, focusing therefore on the purchasing and implementation aspects.

- Expertise in operations research from EUR-ESE. Sustainable solutions require smart decisions, including trade-offs between sustainability, patient outcomes, and costs. This expertise ensures decisions and systems are optimized in this multi-objective setting.

4.c Involvement of partners

The transition to environmentally sustainable healthcare practices is complex and requires input from multiple perspectives to be implemented successfully. At the same time, we notice that this topic receives attention from many actors within and outside of the healthcare sector. We are proud of our societal partners that support this project (Figure 2). They bring in a wide range of expertise and network providing us with many opportunities to make impact beyond scientific publications. These partnerships build upon long-lasting ties between our consortium members and these partners, with a shared history of collaborations within research and educational projects. Our partners will function as co-creators, implementers and as an advisory board for all our activities and research directions, and they will be invited for our monthly webinars to ensure effective knowledge sharing. Apart from the partners that officially join this project, all consortium members have strong ties with societal and industrial partners, government and funding organizations that will benefit the outcomes of this project.

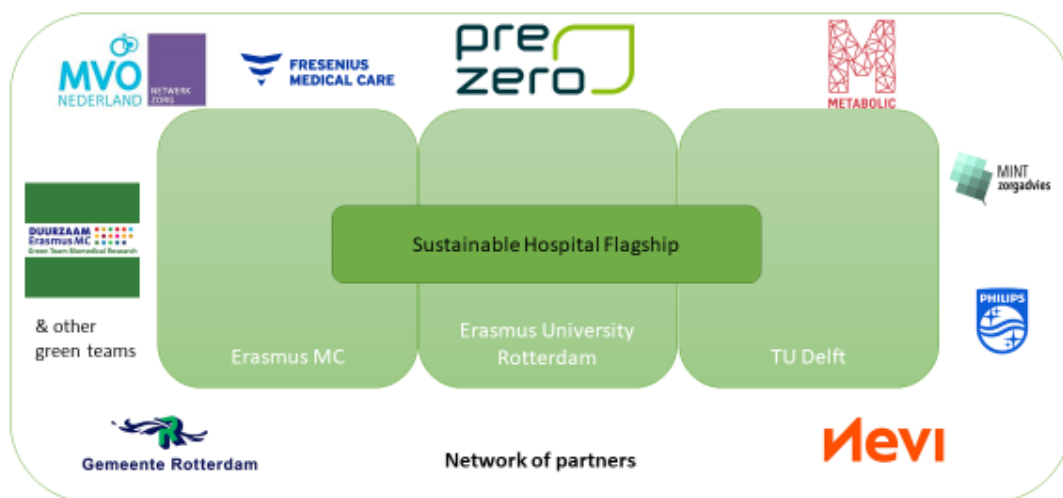


Figure 3: Overview of our partners

4.d Governance, organization and collaboration strategy of the consortium

Effective governance is done by the core team of the consortium that consists of the three leads supported by the community manager/program secretary (WP 7). The community manager will support the use of our physical meet-up and collaboration space, the Sustainable Hospital Field Lab, and the online community that we will build through circle.so, where consortium members, students, partners, green teams, and everyone involved can meet.

The consortium will meet at least biweekly to share updates and insights. The core team will organize a regular webinar for targeted stakeholder engagement. After 2.5 years into the project, WP 6 will start, focusing on integrating all intra-disciplinary and cross-disciplinary insights around the nine (or more) core impact cases. The postdoc (to be hired by this Flagship) will host additional meetings to integrate all research findings into the evidence-based transition model for environmentally sustainable hospitals. This project is very innovative because of its partnership between an academic hospital, a technical university, and business and economics departments, making it possible to innovate within the entire stakeholder range. By focusing strongly on community building in our project setup we facilitate this multidisciplinary way of working whereby we aim to achieve cultural blending and a feeling of joint identity overarching the three joining institutions.

Annexes

I Budget table (including the external funding table)

Budget formulier: Convergentie Flagship Call

Flagship:	Sustainable hospital: transition from waste to value
Flagship acronym:	SHF
Naam Penvoerder:	Nicole Hunfeld
Startdatum en looptijd van het project:	september 2022, 5 jaar
Totaal project budget:	2,8 miljoen

R&D Overview: kostensoorten per type onderzoek (hele project periode)

1.	Loonkosten:								2022		2023		2024		2025		2026		Totaal		Totaal per partner			
	Partner die de kosten maakt	Functie	Aantal fte	Startdatum	Einddatum	Inschaling	Maandbedrag	Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind				
Onderzoeksorganisaties																								
	Erasmus MC	postdoc	1,00	01/09/2022	28/02/2025		€ -	€ 0	€ 88.000,00		€ 92.000,00		€ 48.000,00								€ 228.000,00			
	TU Delft	postdoc	1,00	01/03/2025	31/08/2027		€ -	€ 0					€ 45.000,00		€ 94.000,00		€ 98.090,00				€ 237.090,00			
	Erasmus Universiteit	postdoc	1,00	01/09/2023	31/08/2027		€ -	€ 0	€ 88.000,00		€ 92.000,00		€ 96.000,00		€ 100.000,00		€ 104.280,00				€ 480.280,00			
	Erasmus MC	program secretary	0,50	01/09/2022	31/08/2027		€ -	€ 0	€ 28.750,00		€ 29.750,00		€ 30.750,00		€ 31.750,00		€ 32.990,00				€ 153.990,00			
	Erasmus MC	PhD	1,00	01/09/2022	31/08/2026		€ -	€ 0	€ 50.000,00		€ 58.000,00		€ 63.000,00		€ 69.192,00						€ 240.192,00			
	TU Delft	PhD	1,00	01/09/2022	31/08/2026		€ -	€ 0	€ 50.000,00		€ 58.000,00		€ 63.000,00		€ 69.192,00						€ 240.192,00			
	TU Delft	PhD	1,00	01/09/2023	31/08/2027		€ -	€ 0			€ 50.000,00		€ 58.000,00		€ 63.000,00		€ 69.192,00		€ 69.192,00		€ 240.192,00			
	Erasmus Universiteit	PhD	1,00	01/09/2022	31/08/2026		€ -	€ 0	€ 50.000,00		€ 58.000,00		€ 63.000,00		€ 69.192,00						€ 240.192,00			
	Erasmus Universiteit	PhD	1,00	01/09/2023	31/08/2027		€ -	€ 0			€ 50.000,00		€ 58.000,00		€ 63.000,00		€ 69.192,00		€ 69.192,00		€ 240.192,00			
	Erasmus MC	community manager	0,50	01/09/2022	31/08/2027		€ -	€ 0	€ 28.750,00		€ 29.750,00		€ 30.750,00		€ 31.750,00		€ 32.990,00				€ 153.990,00			
	Erasmus Universiteit	research assistant	3,00	01/09/2022	31/08/2023		€ -	€ 0	€ 184.788,00												€ 184.788,00			
Ondernemingen																								
	Onderneming 1						€ -	€ 0													€ 0,00			
	Onderneming 2						€ -	€ 0													€ 0,00			
	Onderneming 3						€ -	€ 0													€ 0,00			
	Onderneming 4						€ -	€ 0													€ 0,00			
Overige (private) partners																								
	Naam 1						€ -	€ 0													€ 0,00			
	Naam 2						€ -	€ 0													€ 0,00			
	Naam 3						€ -	€ 0													€ 0,00			
	Naam 4						€ -	€ 0													€ 0,00			
																			Totaal	€ 0			Totaal loonkosten	€ 2.639.098

2.	Kosten van materialen en hulpmiddelen:								2022		2023		2024		2025		2026		Totaal		Totaal per partner			
	Partner die de kosten maakt	Materiaal / hulpmiddel						Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind				
Onderzoeksorganisaties																								
	Organisatie 1																				€ 0,00			
	Organisatie 2																				€ 0,00			
	Organisatie 3																				€ 0,00			
	Organisatie 4																				€ 0,00			
Ondernemingen																								
	Onderneming 1																				€ 0,00			
	Onderneming 2																				€ 0,00			
	Onderneming 3																				€ 0,00			
	Onderneming 4																				€ 0,00			
Overige (private) partners																								
	Naam 1																				€ 0,00			
	Naam 2																				€ 0,00			
	Naam 3																				€ 0,00			
	Naam 4																				€ 0,00			
																			Totaal	€ 0			Totaal kosten materialen en hulpmiddelen	€ 0

6.	Totale kosten	€ 59.000	€ 2.876.098
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FINANCIERING - Bijdragen per partner per jaar

	in Euro		2022 in cash in kind		2023 in cash in kind		2024 in cash in kind		2025 in cash in kind		2026 in cash in kind		Totaal in cash in kind	Totaal per partner
Onderzoeksorganisaties														
Convergence			€ 280.000		€ 280.000		€ 280.000		€ 280.000		€ 280.000		€ 1.400.000	€ 1.400.000
1 HORIZON CIRCO			€ 137.500		€ 137.500							€ 275.000	€ 0	€ 275.000
2 Circular Plastic Medical			€ 75.000		€ 75.000							€ 150.000	€ 0	€ 150.000
3 zonMW open						€ 45.000		€ 45.000		€ 45.000		€ 135.000	€ 0	€ 135.000
4 NWA ORC met Aerospace TU Delft						€ 90.000		€ 90.000		€ 90.000		€ 270.000	€ 0	€ 270.000
5 EU Erasmus						€ 25.000		€ 25.000				€ 50.000	€ 0	€ 50.000
Totale bijdrage onderzoeksorganisa-			€ 492.500	€ 0	€ 492.500	€ 0	€ 440.000	€ 0	€ 440.000	€ 0	€ 415.000	€ 0	€ 2.280.000	€ 2.280.000
Ondernemingen														
4 Fresenius Medical				€ 500.000								€ 500.000	€ 0	€ 500.000
5 MVO NL			€ 750		€ 750		€ 750		€ 750		€ 750	€ 0	€ 3.750	€ 3.750
6 Onderneming 3												€ 0	€ 0	€ 0
... Onderneming 4												€ 0	€ 0	€ 0
Totale bijdrage ondernemingen			€ 0	€ 750	€ 500.000	€ 750	€ 0	€ 750	€ 0	€ 750	€ 0	€ 500.000	€ 3.750	€ 503.750
Overige (private) partners														
7 Naam 1												€ 0	€ 0	€ 0
8 Naam 2												€ 0	€ 0	€ 0
9 Naam 3												€ 0	€ 0	€ 0
... Naam 4												€ 0	€ 0	€ 0
Totale overige bijdrage			€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0
PPS-toeslag / subsidies / sponsors														
10 PPS-toeslag												€ 0	€ 0	€ 0
11 <naam>												€ 0	€ 0	€ 0
12 <naam>												€ 0	€ 0	€ 0
...												€ 0	€ 0	€ 0
Totaal subsidies			€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0
Total budget per jaar			in cash € 492.500	in kind € 750	in cash € 992.500	in kind € 750	in cash € 440.000	in kind € 750	in cash € 440.000	in kind € 750	in cash € 415.000	in kind € 750	Totaal incl. PPS-toeslag € 2.783.750	
Funding gap	-													
Total budget			€ 993.250		€ 440.750		€ 440.750		€ 415.750		€ 1.874.750			

II Letters of intent and letters of commitment

On headed paper of the entity. Letters of commitment are preferable to letters of intent and letters of intent are preferable to the absence of these letters.

1. Metabolic

LETTER OF INTENT
for the
Flagship project: Sustainable Hospital

Dear,

I, Pieter van Exter in my capacity of Lead Sustainable Industries at Metabolic hereby confirm that Metabolic intends to contribute to the flagship project **Sustainable Hospital**, as applied for by the main applicant, Dr NGM Hunfeld, flagship lead at Erasmus MC.

Metabolic intends to contribute € [•] in cash towards the project costs in accordance with the budget in the project proposal and budget form.

Metabolic intends to provide an in-kind contribution of sustainability and systems thinking expertise, representing a monetary value of € [•] and further detailed in the project proposal and budget form.

Yours sincerely,



Name:	Pieter van Exter
Position:	Lead Sustainable Industries
Date:	20-03-2022

2. Philips

Van: van Breugel, Bas <bas.van.breugel@philips.com>
Verzonden: vrijdag 25 februari 2022 04:07
Aan: Nicole Hunfeld <n.hunfeld@erasmusmc.nl>
CC: Chaudhari, Sachin <Sachin.Chaudhari@philips.com>
Onderwerp: Re: Proposal Sustainable Hospital: collaboration and letter of intent?

Waarschuwing: Deze e-mail is afkomstig van buiten de organisatie. Klik niet op links en open geen bijlagen, tenzij u de afzender herkent en weet dat de inhoud veilig is.
Caution: This email originated from outside of the organization. Do not click links or open attachments unless you recognize the sender and know the content is safe.

Dear Nicole,

Big thanks for reaching out.
Personally I am very interested to collaborate, learn and improve together.

I will discuss this topic with our business leader for the 'Clinical Applications and Devices' category in Philips.
I expect to be able to come back next week.

Kind regards,

Bas

3. Fresenius Medical Care

Fresenius Medical Care was not able to sign the letter of intent due to complex legal restrictions that take months. However the Medical affairs lead from the Netherlands is invited to pitch our plans for the Sustainable Hospital for the global medical lead of Fresenius Medical Dr Frank Maddux. This will take place in the beginning of April 2022, with the intention that Fresenius Medical Care will fund 2 PhD students.

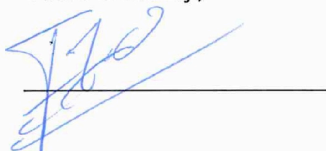
4. PreZero

LETTER OF INTENT
for the
Flagship project: Sustainable Hospital

Dear,

I, Focco Ottens in my capacity of Projectleader Zero waste at Prezero Nederland confirm that Prezero Nederland intends to contribute to the flagship project **Sustainable Hospital**, as applied for by the main applicant, Dr NGM Hunfeld, flagship lead at Erasmus MC.

Yours sincerely,



Name: Focco Ottens

Position: Projectleader Zero waste

Date: March, 21th 2022

5. NEVI

LETTER OF INTENT
for the
Flagship project: Sustainable Hospital

Dear,

I, Karin van IJsselmuide, in my capacity of Manager Maatschappelijke Programma's at Nevi hereby confirm that Nevi B.V. intends to contribute to the flagship project **Sustainable Hospital**, as applied for by the main applicant, Dr NGM Hunfeld, flagship lead at Erasmus MC.

Nevi B.V. intends to contribute € 0,00 in cash towards the project costs in accordance with the budget in the project proposal and budget form.

Nevi B.V. intends to provide an in-kind contribution of knowledge, organisation, facilities and communication, representing a monetary value of € 5.000,00 (50 hrs) and further detailed in the project proposal and budget form.

Yours sincerely,



Name: Karin van IJsselmuide

Position: Manager Maatschappelijke Programma's Nevi

Date: 10 maart 2022

Letter of Intent for a project partner

6. MINT zorgadvies

LETTER OF INTENT
for the
Flagship project: Sustainable Hospital

Dear,

I, Else de Ridder, in my capacity of Founder at MINT zorgadvies hereby confirm that MINT zorgadvies intends to contribute to the flagship project **Sustainable Hospital**, as applied for by the main applicant, Dr NGM Hunfeld, flagship lead at Erasmus MC.

MINT zorgadvies intends to provide an in-kind contribution of supporting the development of online courses on the topic of sustainable healthcare, further detailed in the project proposal.

Yours sincerely,



Name:

Position:

Date:

Letter of Intent for a project partner

7. MVO Nederland

LETTER OF INTENT
for the
Flagship project: Sustainable Hospital

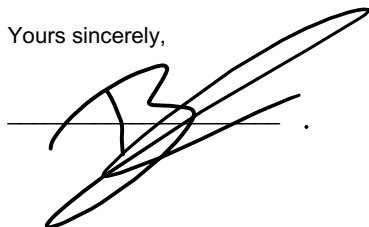
Dear,

I, Bas Maier, in my capacity of Manager Zorg at MVO Nederland hereby confirm that MVO Nederland intends to contribute to the flagship project **Sustainable Hospital**, as applied for by the main applicant, Dr NGM Hunfeld, flagship lead at Erasmus MC.

MVO Nederland intends to contribute € 0,- towards the project costs in accordance with the budget in the project proposal and budget form.

MVO Nederland intends to provide an in-kind contribution of project management, representing a monetary value of € 750,- and further detailed in the project proposal and budget form.

Yours sincerely,



Name: Bas Maier

Position: Sector Zorg

Date: 25-2-2022

Letter of Intent for a project partner

8. Gemeente Rotterdam

LETTER OF SUPPORT
for the
Flagship project: Sustainable Hospital

Dear Nicole Hunfeld,

I, Arno Bonte, in my capacity of alderman of the municipality of Rotterdam hereby confirm that Gemeente Rotterdam supports the flagship project **Sustainable Hospital**, as applied for by the main applicant, Dr NGM Hunfeld, flagship lead at Erasmus MC.

The Gemeente Rotterdam intends to contribute in kind by dissemination of knowledge and best practices resulting of this flagship through their network of hospitals and healthcare providers in the Rotterdam area, specifically through sharing knowledge through the Rotterdam Klimaattafel Healthcare.

Yours sincerely,

Name: Arno Bonte

Position: Vice Mayor for Sustainability, Clean Air and Energy Transition of the City of Rotterdam

Date: 15/03/2022

III Letter of support from department head of the main lead

 Erasmus MC Universitair Medisch Centrum Rotterdam	Intensive Care Volwassenen
Concerning: Letter of intent	Doorkiesnummer +31-10-7033629 Faxnummer Kamernummer Na-409 E-mail d.gommers@erasmusmc.nl Ons kenmerk 1701003 008-22 Datum February 25th, 2022
Dear sir/madam,	Postal address Postbus 2040 3000 CA Rotterdam Street address Dr. Molewaterplein 40 3115 GD Rotterdam
I, prof.dr. Diederik Gommers, head of the department of Intensive Care (Erasmus MC, Rotterdam), intend to contribute to the flagship proposal Sustainable Hospital, transition from linear to circular, as applied for by Dr Nicole Hunfeld, corresponding lead of this flagship. Given the importance of this topic and its impact on climate change and health, and the strength of the convergence consortium, I'm fully supporting this Flagship application and hope for a positive decision on this ground breaking proposal with high societal impact	Parking Wylemaweg 12 / Westzeedijk 361 Head of Department Prof.dr.D. Gommers Substitute head of Department Dr. J. van Bommel Trainer Dr. H. Endeman Head of CTB R.J.M. van den Biggelaar
Yours sincerely,	Staff J.P.C. van den Akker Prof.dr. J. Bakker K. Bokhoven J. Bunge Dr. D. Ros Reis Miranda Dr. J.L. Epker P.C. Geritsen Dr. H.R.H. de Geus M.W. de Groot J.E. de Haan Prof.dr. L. Heunks B. van der Hoven Dr. N.G.M. Hunfeld Prof.dr. C. Ince Dr. M. van der Jagt L.S.D. Jewballi S.C.J. van der Kleij Dr. E. Klijn D.B.M. Kock-Cordeiro Dr. E.J.O. Kompanje J.H.J. Meeder C.L. Meuwese G. Prins-van Gilst Dr. A. Struijs R.J. van Thiel M. Türk Dr. C.A. den Uijl J.D. Weigel
	
Prof.dr. D.A.M.P.J. Gommers Head of Department Intensive Care/CTB Erasmus MC	
	www.erasmusmc.nl

IV References

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